

Minor Project Report

Project Title:

Exploratory Data Analysis on Diwali Sales Dataset

Project Objective:

The aim of this project is to perform a detailed Exploratory Data Analysis (EDA) on the Diwali Sales dataset to gain meaningful insights into consumer buying behavior, product performance, and regional sales patterns.

Tools and Libraries Used:

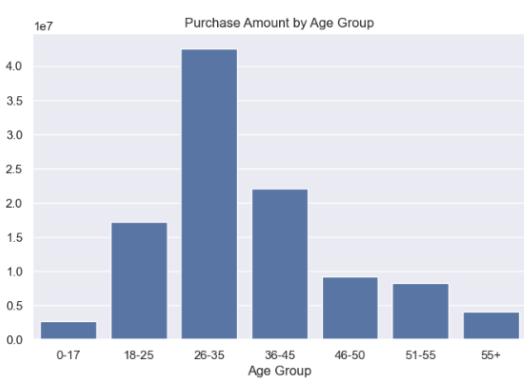
Python , Jupyter Notebook , Pandas , NumPy , Matplotlib , Seaborn.

Screenshots of Key Visualizations:

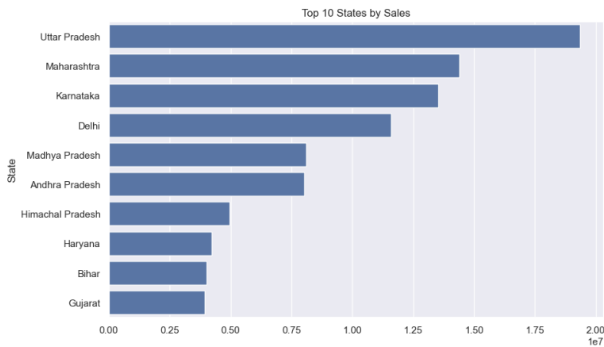
1)



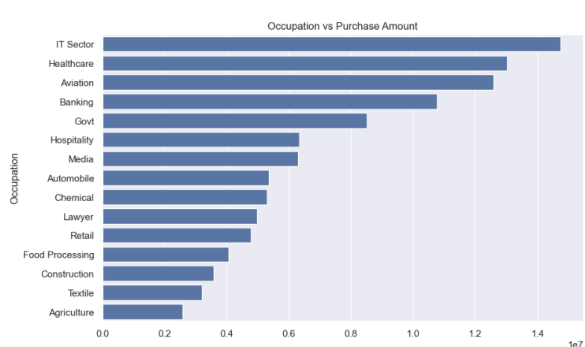
2)



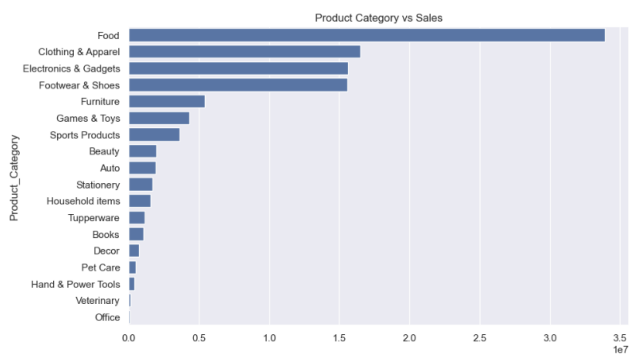
3)



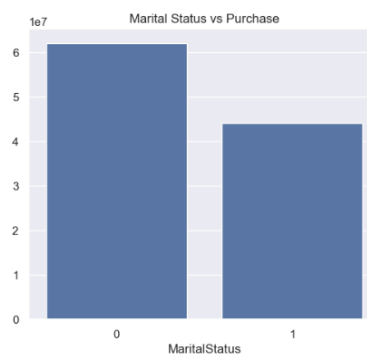
4)



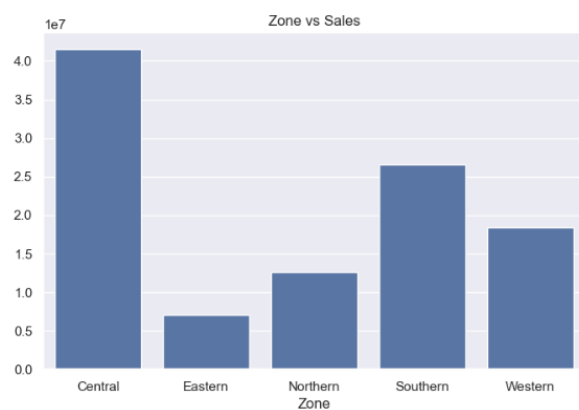
5)



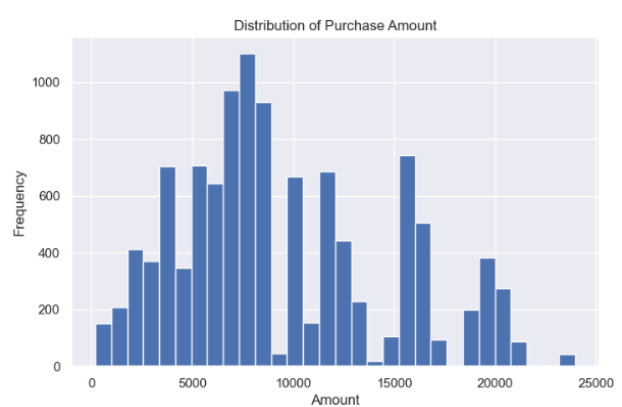
6)



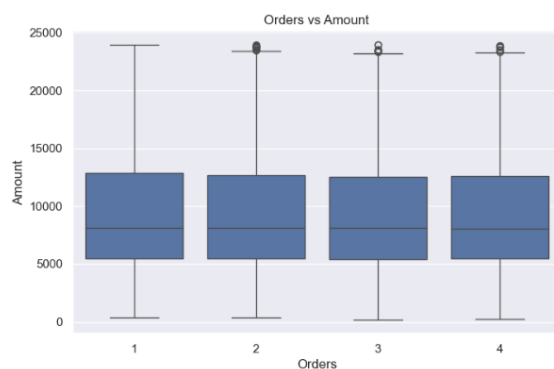
7)



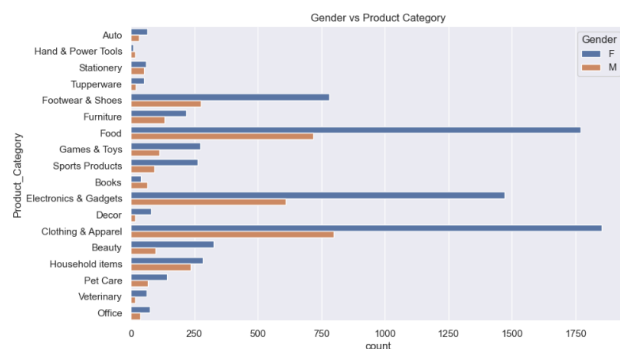
8)



9)



10)



Summary of Insights:

- 1) Female customers contribute significantly more to total sales compared to male customers, making them the dominant buyer group during Diwali.
- 2) The age group 26–35 accounts for the highest purchase amount, indicating that young working professionals are the most active consumers.
- 3) Uttar Pradesh, Maharashtra, and Karnataka are the top contributing states, showing strong regional sales concentration in these areas.

- 4) Customers working in the IT, Healthcare, and Aviation sectors demonstrate the highest spending behavior among all occupations.
- 5) Food, Clothing & Apparel, and Electronics are the most popular product categories, indicating high demand for festive essentials and lifestyle products.
- 6) Unmarried customers show higher overall purchase amounts compared to married customers, suggesting stronger spending trends among unmarried individuals.
- 7) The Central and Southern zones generate the highest total sales, highlighting these regions as key markets for Diwali purchases.
- 8) Most purchase amounts fall within a moderate range, indicating that the majority of customers are mid-level spenders rather than extreme high-value buyers.
- 9) There is no strong direct relationship between number of orders and purchase amount, suggesting customers often buy fewer but higher-value products.
- 10) Female customers dominate purchases across most product categories, particularly in Clothing and Food, reinforcing their key role in festive shopping.

Challenges Faced and Solutions:

In the course of carrying out the Exploratory Data Analysis, a number of challenges were faced in working with the Diwali Sales dataset. The first of these was dealing with the null values in the dataset. This was addressed by removing the null values using the `dropna()` function. The second was dealing with the encoding error during the import of the dataset. This was addressed by specifying the correct encoding format, which in this case was the 'unicode_escape' format, in the `read_csv()` function. The third was dealing with the incorrect data types in some of the columns of the dataset. This was addressed by changing the data types of these columns to the correct types. Challenges in creating the visualizations, such as the overlapping of labels, were addressed by changing the figure size, rotating the labels, and choosing the correct visualization types. These solutions ensured clean, accurate, and meaningful analysis of the dataset.

Conclusion:

The Exploratory Data Analysis of the Diwali Sales dataset provided significant insights into the behaviour of customers, the performance of products, and the sales trends in different regions. The analysis showed that female customers and customers in the range of 26 to 35 years of age are the most significant contributors to the sales. The states such as Uttar Pradesh and the zones such as the Central region have significant sales performance. The most popular product categories such as Food, Clothing, and Electronics have significant contributions to the sales. The analysis helps in understanding the behavior of customers, and businesses can take significant steps to enhance their marketing strategies, product offerings, and sales performance during festive seasons.